Senior Project Outline

2nd version (current)

Konrad Owczarek

Overview

I will be designing, making, and publishing a website, as well as a mobile application for Android (and iOS, if time permits) that aims to make a convenient and easy-to-use shopping list for the user, with some additional features. Simplicity is key in this shopping list application – there should be no need for excessive instructions, and it should have a very simple user interface. I will be focusing on implementing many of the software design principles from CSC 307, focusing on having a good software development cycle. At the conclusion of the project, I will publish the mobile app on the Android Play Store (and maybe the iOS App Store) and deploy the website.

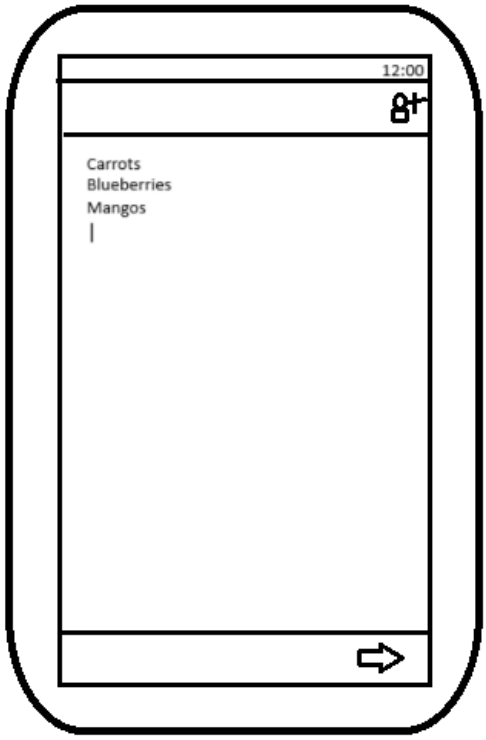
Specifics

The app and website will be focused on a simple and intuitive design. It will be able to take input, which will either be items that the user wants to buy, or a more detailed recipe/recipe description, or parts that the user needs for something, which may need to be parsed by a recipe, parts, AI, or other API. This will then potentially be expanded into the list of items the user should buy. The items themselves can be groceries, or other items, such as home goods. The app and website will attempt to locate the best deal, both money-wise and timewise, for the user. All the while, it should be making sure that all is in line with the health and dietary preferences the user may or may not have inputted. Additionally, the lists should be sharable with other people, who may or may not be users of the app, for a seamless and account-free experience.

Technologies/Methods

The app will be created using software to create Android applications (such as Android Studio), and the website will be created using the tools from CSC 307, using technologies like node.js. If I end up making the iOS app, I will find suitable software for that. For parsing recipes as well as user inputs, I will likely use a combination of regular programming and an API for recipes, parts, AI, or something else; for example, the Edamame API for recipes. To notify the user of nearby stores that offer the necessary products, I will most likely use a hardcoded list of items stores generally have, but if more reliable options become

available, I will use those instead. I will be implementing basic technologies such as prettier and Jest or their equivalents in the language I use. I will also be creating some sort of personal backend API for ease of connectivity between all platforms and a more seamless development experience. Finally, I will be using some kind of database, such as MongoDB, and ensure the security of the data with encryption.

Potential upgrades

I may initially focus on SLO for a higher-quality database but also may have a more general database for the rest of the US – adding this general database would be a potential upgrade. A direct partnership with stores such as Target, allowing for real-time product information, could be a great improvement. Another improvement could be a connection with APIs such as Instacart, allowing the grocery list to be automatically sent to an Instacart worker as if the user had selected the items in the application. Additionally, having detailed nutritional information shown could be useful for health-conscious individuals. A final improvement could be the introduction of a map or instructions as to where to go in the store for the items.